

RADIAL ARCHED BEAM DOME SYSTEM



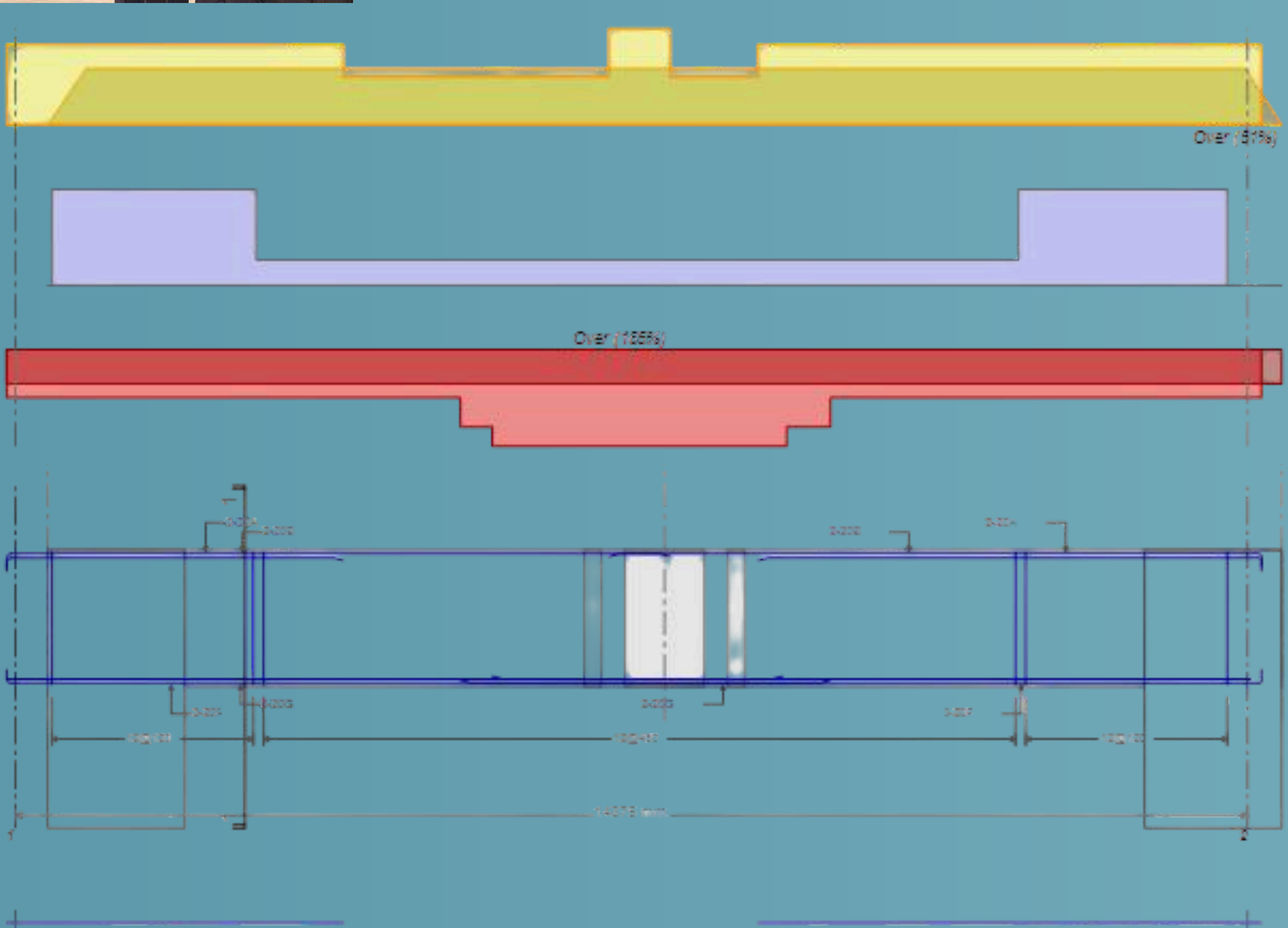
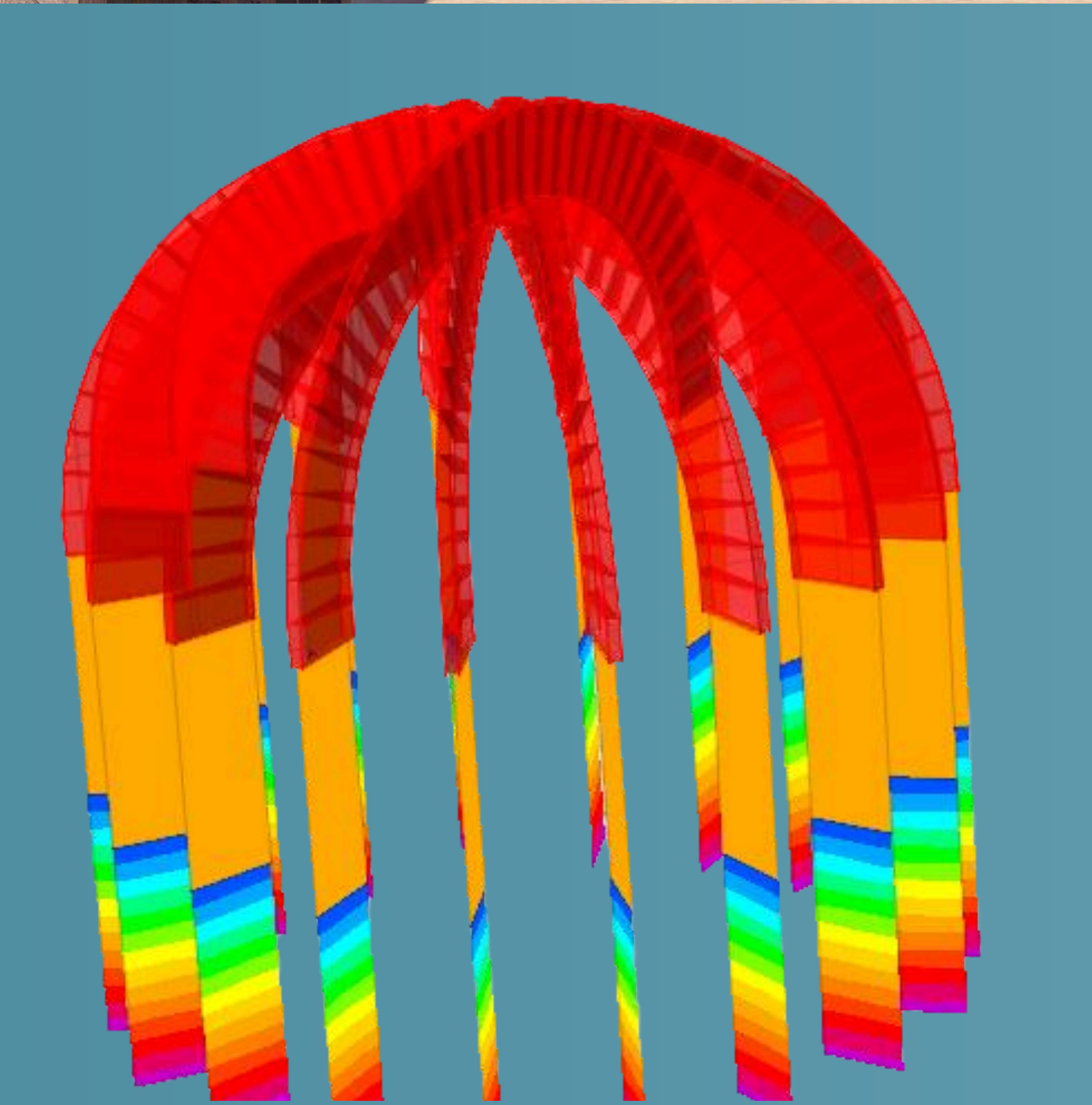
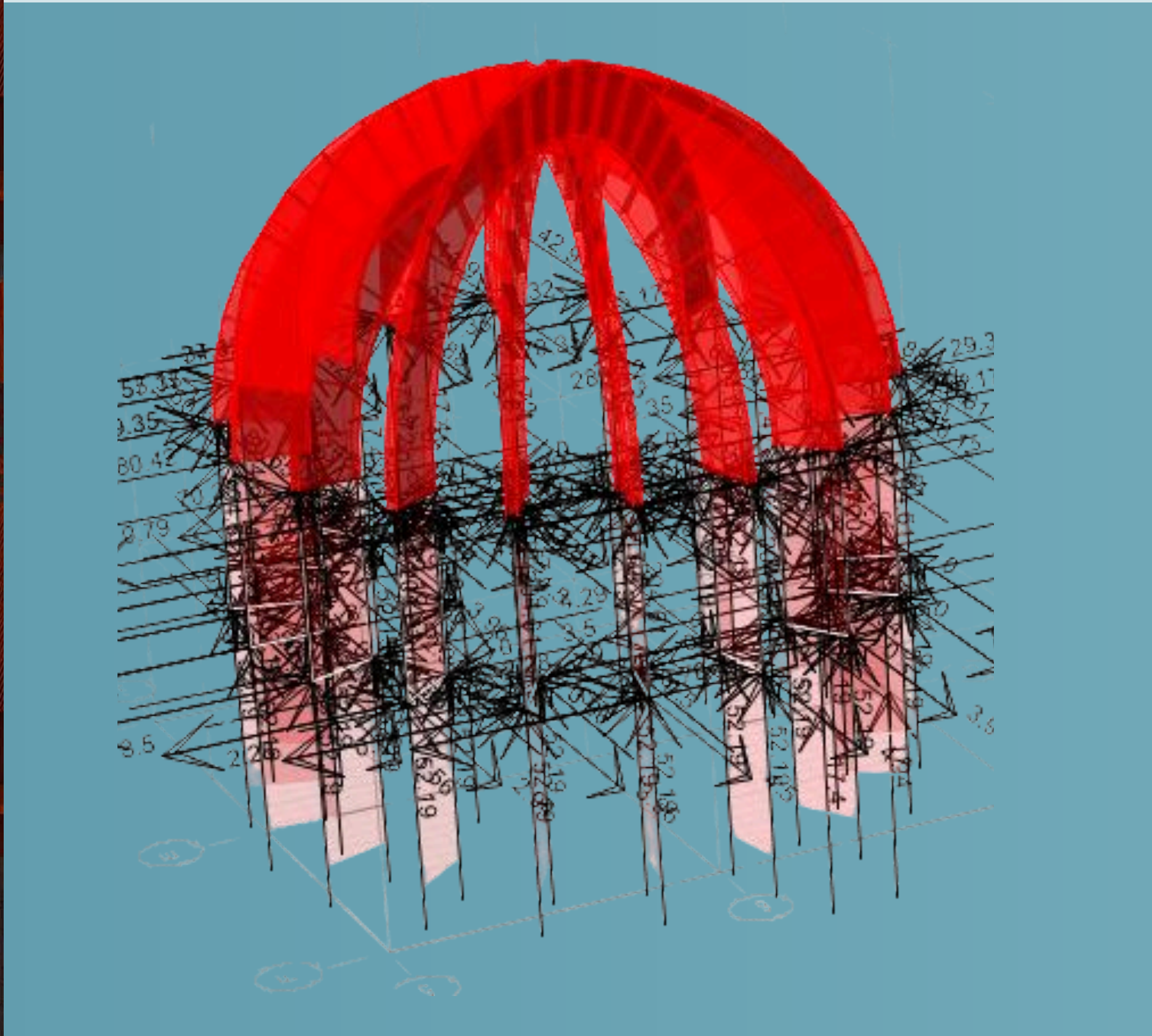
DESIGN METHODOLOGY

- MODELING: SINGLE ARCH RIB MODELED VIA ARC (3 POINTS) TOOL IN ELEVATION.
- GENERATION: RADIAL REPLICATION (\$360^\circ\$/CIRC\$) USED TO CREATE THE DOME SKELETON AND ENSURE GEOMETRIC SYMMETRY.
- ANALYSIS: STRUCTURAL INTEGRITY VERIFIED THROUGH CSI ETABS FINITE ELEMENT ANALYSIS (FEA).
- COMPLIANCE: ALL FRAME ELEMENTS DESIGNED TO SBC 304 (SAUDI BUILDING CODE) STANDARDS.

Element	Reinforcement Detail
Main Bars	5 Φ 20 Top & Bottom (Symmetrical Compression Design)
Shear (Ends)	Φ 12 @ 120mm (High-shear zones)
Shear (Mid)	12 Φ @ 450mm (Standard zones)
Durability	Optimized concrete cover (50mm)for "Exposed to Air" conditions per SBC.

DEFORMATION & STABILITY PERFORMANCE

- ARCH ACTION: CONVERTS VERTICAL GRAVITY LOADS INTO AXIAL COMPRESSION, DRASTICALLY REDUCING MID-SPAN DEFLECTION.
- HOOP STIFFNESS: RADIAL SYMMETRY GENERATES "HOOP STRESS" THAT NEUTRALIZES HORIZONTAL THRUST.
- WALL SUPPORT: EVEN LOAD DISTRIBUTION ACROSS THE PERIMETER MINIMIZES OUTWARD DISPLACEMENT AND WALL DEFORMATION.
- RESULT: HIGH-STIFFNESS SYSTEM WITH MINIMAL DEFORMATION UNDER MAXIMUM DESIGN LOADS.



ETABS

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All concrete frames passed the design check.

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STRUCTURAL SYSTEM SELECTION

System	Description	Advantages	Disadvantages	Suitability for Your Project
Flat Plate	Uniformly thick slab directly on columns.	Thinnest system; maximum clear height for MEP.	Low punching shear capacity; struggles with high loads (> 3kN/m^2).	Unsuitable. Would fail at columns due to hospital live loads and garden areas.
Flat Slab (With Drop Panels)	Slab with thickened area (drop panels) at columns.	Excellent punching shear resistance; handles high live loads (4.79 kN/m^2); clear ceiling for ducts between drops.	Drop panels (40cm) require careful MEP coordination near columns.	SELECTED. Best compromise for heavy hospital loads and duct clearance.
Ribbed / Waffle Slab	Slab with a grid of ribs (joists).	Lightest weight; excellent for very long spans.	High MEP interference; ribs block horizontal pipe and duct runs.	Unsuitable. Crashes with the large ducts required for a hospital.
Solid Slab (Beam & Slab)	Slab supported by a primary and secondary beam grid.	Highest structural stiffness and load capacity.	Lowest MEP clearance; beams force ducts to drop lower, reducing ceiling height.	Unsuitable. User constraint prohibits beams due to service clashes.

